

of 0.1763 MAE. The largest current-gain interaction is 0.000443 MAE.

Initialization recovery exposes a separate boundary. Five perturbations retain mean persistent recovery-or-censor times between 1.18 and 1.41 h, close to the nominal 1.20 h. Scaling the resistance surfaces by -10% increases the equal-weight mean to 5.10 h because one low-load holdout trajectory remains outside the recovery band for the 24-h horizon. The analysis supports local stability of most measurement-disturbance responses but does not establish lookup independence of baseline accuracy or observer re-anchoring.

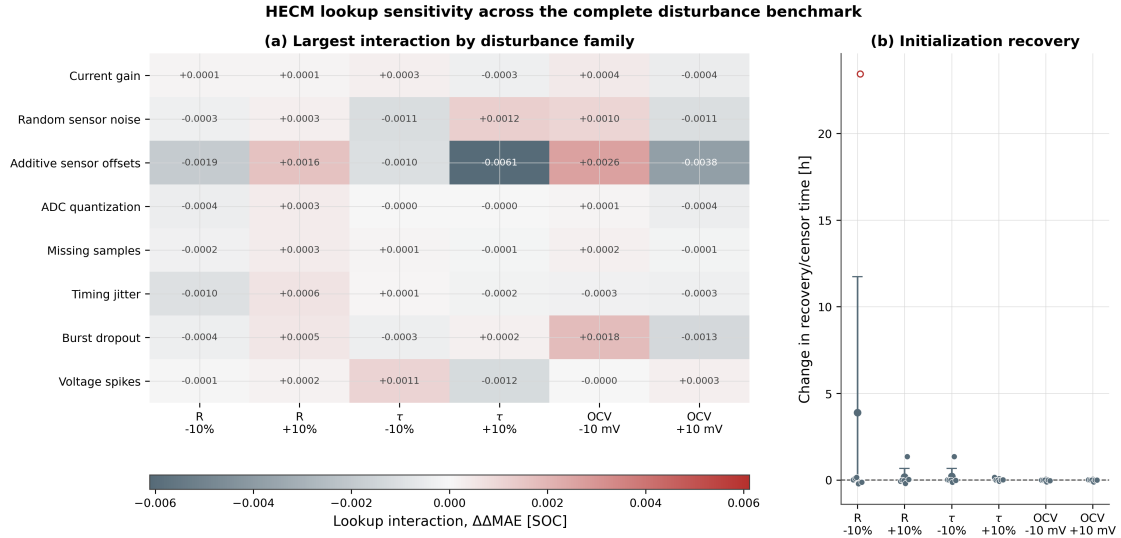


Figure A.3: HECM lookup sensitivity across the complete disturbance benchmark. Panel (a) reports the largest lookup interaction in each robustness family under local resistance, time-constant, and OCV perturbations. Panel (b) reports the corresponding change in persistent initialization recovery-or-censor time. Open red points identify cell means containing a censored recovery window.